Pranay Singh, EIT

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Biosketch

Pranay Singh is a Ph.D. candidate in Civil and Structural Engineering at the University at Buffalo. His research focuses on strain-hardening cementitious composites (SHCC/ECC), 3D concrete printing, and the experimental and numerical assessment of prestressed concrete bridge girders. His doctoral work examines crack control in prestress-transfer regions and flexural behavior of girders with unbonded and hybrid prestressing systems.

Prior to joining the University at Buffalo in 2023, he worked as a Senior Project Scientist at IIT Delhi on seismic isolation systems and as a Project Engineer at the National Council for Cement and Building Materials on advanced concrete and structural testing. He earned his M.Tech. from IIT Patna in 2021, where he received the Institute Silver Medal.

Education

University at Buffalo, State University of New York Aug 2023 – May 2027 (expected)
Ph.D. in Civil and Structural Engineering *Buffalo, New York*

Indian Institute of Technology (IIT) Patna Aug 2019 – Jul 2021
M.Tech. in Civil Engineering *Patna, India*

Bhagalpur College of Engineering Aug 2015 – Jul 2019
B.Tech. in Civil Engineering *Bhagalpur, India*

Professional Licensure & Examinations

Fundamentals of Engineering (FE) Exam *Passed*
Principles and Practice of Engineering (PE) Exam *Scheduled*

Ph.D. Dissertation

Strain Hardening Cementitious Composites for Prestressed Concrete Bridge Girders

Advisors: Prof. Ravi Ranade, and Prof. Pinar Okumus

Support: Transportation Infrastructure Precast Innovation Center (TRANS-IPIC), Tier 1 University Transportation Center.

Approach: 3D-printable SHCC/ECC development, mechanical and rheological characterization, prestressed girder experiments, and nonlinear finite element analysis to improve end-zone crack control and flexural behavior with unbonded and hybrid prestressing.

Research & Professional Experience

University at Buffalo, State University of New York Aug 2023 – Present
Doctoral Researcher, Institute of Bridge Engineering *Buffalo, New York*

- Developing and characterizing 3D-printable SHCC/ECC for crack control in pretensioned bridge girder end zones.
- Evaluating conventional-concrete and ECC beams with unbonded and hybrid prestressing through four-point bending tests and nonlinear ABAQUS analysis.

Indian Institute of Technology (IIT) Delhi Aug 2022 – Jul 2023
Senior Project Scientist *New Delhi, India*

- Developed and analyzed a self-centering, friction-based sliding base-isolation concept using nonlinear earthquake-response simulations.
- Evaluated isolation-system design parameters and structural-control effectiveness under seismic loading.

National Council for Cement and Building Materials (NCCBM) Aug 2021 – Aug 2022
Project Engineer *Ballabgarh, Faridabad, India*

- Investigated ultra-high-performance, fiber-reinforced, and geopolymer concretes using mechanical, fracture, flexural, and shear testing.

- Supported research on roller-compacted concrete and thermal monitoring and control of mass concrete in construction.

Selected Research Projects

ECC Girders with Unbonded and Hybrid Prestressing | *University at Buffalo*

- Compared the influence of tensile strain-hardening on cracking, flexural resistance, deformation, and strand stress development using beam tests and calibrated numerical models.

3D-Printed SHCC for Prestressed Girder End Zones | *University at Buffalo*

- Developed printable SHCC mixes and permanent cover concepts for prestressed girder ends; assessed fresh-state printability, tensile performance, and prestress-transfer cracking.

Steel-Wool-Reinforced Concrete for Crack Mitigation | *University at Buffalo*

- Characterized tensile fracture and residual post-cracking resistance of steel-wool-reinforced concrete for crack-control applications.

Seismic Isolation and Shape Memory Alloy Dampers | *IIT Delhi / IIT Patna*

- Numerically assessed sliding isolation strategies and SMA-based dampers to reduce seismic response and pounding between adjacent buildings.

Advanced and Sustainable Concrete Materials | *NCCBM*

- Developed and characterized UHPC, high-strength fiber-reinforced concrete, and alkali-activated geopolymer concrete; studied ductility and flexural, shear, and fracture performance.
- Additional projects: roller-compacted concrete for dams; field temperature control and monitoring of mass concrete.

Publications

Google Scholar Profile

Journal Articles

1. **Pranay Singh**, P. N. Ojha, Brijesh Singh, and Abhishek Singh. “Crack detection and crack segmentation in concrete beams undergoing mode I fracture using computer vision and convolutional neural network.” *Canadian Journal of Civil Engineering*, 50(5), 432–443, 2023.
2. **Pranay Singh**, Brijesh Singh, P. N. Ojha, Amit Trivedi, Abhishek Singh, and Chirag Pede. “Fracture behavior of slag & fly ash-based geopolymer concrete in comparison with OPC-based conventional concrete, including the effect of steel and hybrid fibers.” *Revista Ingeniería de Construcción*, 38(2), 2023.
3. Sourav Gur, Koushik Roy, and **Pranay Singh**. “Seismic performance assessment of adjacent building structures connected with superelastic shape memory alloy damper and comparison with yield damper.” *Structural Control and Health Monitoring*, e2926, 2022.
4. P. N. Ojha, Sumit Kumar, Puneet Kaura, Brijesh Singh, and **Pranay Singh**. “Experimental investigation on effect of corrosion on curvature-ductility relationship of RCC member in flexure.” *Journal of Building Materials and Structures*, 9(1), 74–86, 2022.
5. P. N. Ojha, Amit Sagar, Brijesh Singh, Harish Sharma, and **Pranay Singh**. “Temperature Control and Monitoring for Mass Concrete in Raft under Hot Weather Condition – An Indian Case Study.” *Sustainable Structures and Materials*, 4(2), 19–28, 2022.
6. P. N. Ojha, Brijesh Singh, Shivam Prakash, **Pranay Singh**, Manish K. Mandre, and Suresh Kumar. “Effect of high ratio fly ash on roller compacted concrete for dam construction.” *Research on Engineering Structures and Materials*, 8(2), 233–251, 2022.
7. P. N. Ojha, Brijesh Singh, and **Pranay Singh**. “Empirical equations for prediction of split tensile and flexural strength of high strength concrete including effect of steel fiber.” *Research on Engineering Structures and Materials*, 2022.
8. P. N. Ojha, Brijesh Singh, **Pranay Singh**, Abhishek Singh, and Manish Kumar Mandre. “Study on effect of fly ash and limestone powder on compressive strength of roller compacted concrete for dam construction.” *Journal of Asian Concrete Federation*, 8(1), 37–51, 2022.
9. P. N. Ojha, Brijesh Singh, Abhishek Singh, and **Pranay Singh**. “Fracture behavior of plain and fiber-reinforced high strength concrete containing high strength steel fiber.” *Research on Engineering Structures and Materials*, 8(3), 583–602, 2022.

10. P. N. Ojha, Brijesh Singh, Amit Trivedi, **Pranay Singh**, Abhishek Singh, and Chirag Pede. "Short-term mechanical performance and flexural behavior of reinforced slag-fly ash-based geopolymer concrete beams in comparison to OPC-based concrete beams." *Research on Engineering Structures and Materials*, 9(1), 31–51, 2022.
11. P. N. Ojha, **Pranay Singh**, Brijesh Singh, and Abhishek Singh. "Fracture behavior of hybrid fiber reinforced normal strength and high strength concrete: comparison with plain and steel fiber reinforced concrete." *Research on Engineering Structures and Materials*, 9(1), 243–262, 2022.
12. P. N. Ojha, **Pranay Singh**, Brijesh Singh, Abhishek Singh, Ajay, and Amit Sagar. "Experimental and FEM analysis for fracture performance evaluation of concrete made with recycled construction and demolition waste aggregates." *Periodica Polytechnica Civil Engineering*, 2022.
13. P. N. Ojha, **Pranay Singh**, Brijesh Singh, Abhishek Singh, Amit Sagar, and Ravi Yadav. "Statistical characteristics of compressive strength of normal & high strength concrete and concrete made with recycled aggregate." *Revista Ingeniería de Construcción*, 37(3), 2022.
14. Brijesh Singh, **Pranay Singh**, P. N. Ojha, and Abhishek Singh. "Experimental study on stress-strain characteristics of ultra high strength concrete and its effect on stress block parameters for flexural design of building." *Research on Engineering Structures and Materials*, 9(2), 503–525, 2022.
15. **Pranay Singh**, P. N. Ojha, Brijesh Singh, Piyush Mittal, and Abhishek Singh. "Experimental determination of stress-strain characteristics of very high strength unconfined concrete in compression including comparison with normal and high strength concrete." *Revista Ingeniería de Construcción*, 37(3), 2022.
16. P. N. Ojha, **Pranay Singh**, Brijesh Singh, Abhishek Singh, Amit Prakash, and Amit Sagar. "Stress-strain characteristics and energy absorption analysis of construction and demolition waste recycled aggregate concrete in compression." *Sustainable Structures and Materials: An International Journal*, 4(2), 29–43, 2021.

Manuscripts in Preparation *(working titles)*

1. "Performance-Driven Development of 3D-Printable Strain-Hardening Cementitious Composites for Prestressed Concrete Applications." *In preparation*.
2. "Interface Mechanics of 3D-Printable Strain-Hardening Cementitious Composites." *In preparation*.
3. "Flexural Performance of ECC Girders with Unbonded and Hybrid Prestressing: Experiments and Nonlinear Finite Element Analysis." *In preparation*.

Technical Reports

1. **Pranay Singh**, Ravi Ranade, Chi Zhou, and Pinar Okumus. *3D Printed Advanced Materials to Mitigate Prestressed Concrete Girder End Cracks*. Transportation Infrastructure Precast Innovation Center (TRANS-IPIC), Tier-1 University Transportation Center, U.S. Department of Transportation, Report No. UB-23-RP-02, 2025.

Conference Papers & Proceedings

1. **Pranay Singh**, Chi Zhou, Pinar Okumus, and Ravi Ranade. "Development of 3D Printable Strain-Hardening Cementitious Composites for Bridge-Related Applications." *XI International Symposium on Fiber Reinforced Concrete (BEFIB 2024)*, Dresden, Germany, 2024.
2. **Pranay Singh**, Venkateswara Swamy Gadde, Chi Zhou, Pinar Okumus, and Ravi Ranade. "3D Printed Advanced Materials to Mitigate Prestressed Concrete Girder End Cracks." *103rd Transportation Research Board (TRB) Annual Meeting*, Washington, D.C., 2024.
3. **Pranay Singh**, Sourav Gur, and Koushik Roy. "Seismic Performance of Coupled Buildings Connected by Yield and SMA Dampers." *ASPS Conference Proceedings*, Vol. 1(4), 1129–1137, 2022.
4. Amit Trivedi, Brijesh Singh, Abhishek Singh, P. N. Ojha, **Pranay Singh**, and Dinesh Kumar. "Shear behaviour of reinforced alkali activated slag and fly ash concrete under ambient curing: comparison with OPC-based concrete." *17th NCB International Conference on Cement*, *Concrete and Building Materials*, 2022.
5. Sourav Gur, **Pranay Singh**, and Koushik Roy. "Seismic response of adjacent building structure connected with superelastic damper: comparison with yield damper." *EURODYN 2020 – XI International Conference on Structural Dynamics*, 4696–4709, 2020.

Other Publications

1. Abhishek Singh, Brijesh Singh, P. N. Ojha, Amit Trivedi, and **Pranay Singh**. "Flexural and Fracture Performance of Geopolymer Concrete Cured at Ambient Condition." 2022.

Presentations & Talks

105th TRB Annual Meeting | *Washington, D.C.* | Jan 2026 | *Enhancing Anchorage Zone Durability in Pretensioned Girders Using Additively Manufactured SHCC Shells: Material Design and Proof-of-Concept Study* – Poster Presentation

ACI Concrete Convention – Fall 2025 | *Baltimore, MD* | Oct 2025 | *Printing Meets Precast: 3D-Printed Ductile Concrete Covers for Durable Bridge Girder Ends* – Poster Presentation

ASU–Swinburne Workshop on Advances in Concrete 3D Printing | *Hawthorn, Australia* | Jun 2025 | *3D Printing of Advanced Cementitious Materials for Bridge Applications* – Poster Presentation

TRANS-IPIC Annual Workshop | *Rosemont, IL* | Apr 2025 | *3D Printed Advanced Materials to Mitigate Prestressed Concrete Girder End Cracks – Phase II* – Research Presentation

TRANS-IPIC Research Highlights Webinar | *Virtual* | Feb 2025 | *3D Printed Advanced Materials to Mitigate Prestressed Concrete Girder End Cracks* – Research Presentation

104th TRB Annual Meeting | *Washington, D.C.* | Jan 2025 | *Additive Manufacturing of Concrete for Corrosion Protection of Prestressed Concrete Bridges* – Research Presentation

AAMCT Conference | *Irvine, CA* | Nov 2024 | *Development of 3D Printable Strain-Hardening Cementitious Composites for Bridge-Related Applications* – Research Presentation

BEFIB 2024 | *Dresden, Germany* | Sep 2024 | *Development of 3D Printable Strain-Hardening Cementitious Composites for Bridge-Related Applications* – Research Presentation

TU Dresden Summer School (GRK 2250) | *Dresden, Germany* | Sep 2024 | *3D-Printed Strain-Hardening Cementitious Composites as Permanent Formwork for Pretensioned Concrete Girders* – Poster Presentation

TRANS-IPIC Research Webinar | *Virtual* | Jul 2024 | *3D Printed Advanced Materials to Mitigate Prestressed Concrete Girder End Cracks* – Research Presentation

TRANS-IPIC Annual Workshop | *Rosemont, IL* | Apr 2024 | *Development of 3D Printable Strain-Hardening Cementitious Composite for Bridge-Related Applications* – Research Presentation

103rd TRB Annual Meeting | *Washington, D.C.* | Jan 2024 | *3D Printed Advanced Materials to Mitigate Prestressed Concrete Girder End Cracks – Material Properties Identification and Mix Tailoring* – Poster Presentation

Honors, Awards & Fellowships

American Concrete Institute (ACI) Travel Stipend | ACI Concrete Convention, Fall 2025	2025
ASU–Swinburne Workshop Travel Grant | Advances in Concrete 3D Printing	2025
Graduate Assistantship, Institute of Bridge Engineering | University at Buffalo	2023–2024
Institute Silver Medal | IIT Patna – Highest CGPA in Civil & Environmental Engineering	2021
MHRD Fellowship, Government of India | Master's studies at IIT Patna	2019–2021

Academic & Professional Service

Journal Reviewer | *Automation in Construction* (Elsevier); *Construction and Building Materials* (Elsevier); *Journal of Building Engineering* (Elsevier); *Transportation Research Record* (SAGE Publications); *Journal of Materials in Civil Engineering* (American Society of Civil Engineers, ASCE); *Journal of Structural Integrity and Maintenance* (Taylor & Francis)

Conference Reviewer | Transportation Research Board (TRB) Annual Meeting; Conference on Advancing Additive Manufacturing and Construction in Transportation (AAMCT); Digital Concrete (3DPCMS) Conference

Scientific Committee Member | Digital Concrete (3DPCMS) Conference, Nanjing, China | 2026

Session Moderator | Conference on Advancing Additive Manufacturing and Construction in Transportation (AAMCT), Irvine, CA | 2024

Professional Memberships

American Concrete Institute (ACI) • RILEM • American Society of Civil Engineers (ASCE) • Precast/Prestressed Concrete Institute (PCI) • fib

Technical Skills

Structural & Bridge Engineering: Prestressed and Reinforced Concrete, Bridge Girder Analysis, Nonlinear Finite Element Analysis (ABAQUS), Structural Dynamics, Structural Control

Advanced Concrete Materials: ECC/SHCC, Fiber-Reinforced Concrete, UHPC, Rheology, Mechanical Testing, Fracture and Post-Cracking Characterization

Additive Manufacturing: 3D Concrete Printing, Mix Design, Printability, Fresh-State Characterization

Scientific Computing & Data Analysis: Python, MATLAB, NumPy, Pandas, SciPy, Data Processing and Visualization

Machine Learning: Regression, Classification, Decision Trees, Random Forests, Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN)
High-Performance Computing: Linux, SLURM, Parallel Computing